

EXERCISE I

In reference to the Strassen Matrix Multiplication technique

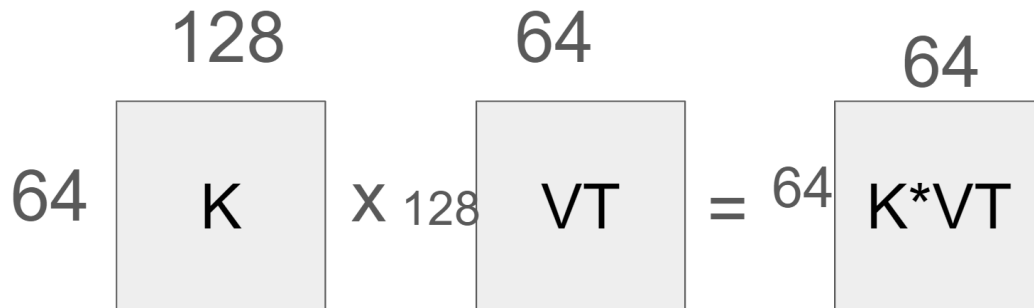
I.a) Show the calculation of the product of the terms P1-P7 for the matrix product $[[1,2][3,4]] \times [[5,6][7,8]]$

I.b) estimate the number multiplications and additions needed to compute the product of two 4x4 matrices with the Strassen method (use the recursive formula)

I.c) Using the asymptotic approximation compute the number of operations for the same matrix product of (I.b). Explain why there are differences, if any.

EXERCISE II

Consider a simplified attention module $A = \text{SoftMax}(K \cdot VT)$, where K is 64×128 matrix and VT is 128×64 . Matrix elements in FP16 format



II.a) Computed the total memory space needed to store, K,VT,A

II.b) Compute the total number of load and store operations (read and writes) needed for computing the attention module, assuming that matrix multiplication is computed with the trivial algorithms (triple nested loop iteration)

MatMul:

w/ psum in memory: $64 \cdot 64 \cdot 128 \cdot 3$ LOAD + $64 \cdot 64 \cdot 128$ STORE = 1.572.864 LOAD + 524.288 STORE

w/ psum in local reg.: $64 \cdot 64 \cdot 128 \cdot 2$ LOAD + $64 \cdot 64$ STORE = 1.048.576 LOAD + 4096 STORE

Softmax:

Per row:

Denominator: 64 LOAD

Normalization: 64 LOAD + 64 STORE

Total Softmax: $64 \cdot (\text{denominator} + \text{normalization}) = 64 \cdot (64 \text{ LOAD} + 64 \text{ LOAD} + 64 \text{ STORE}) = 8192 \text{ LOAD} + 4096 \text{ STORE}$

Total Attention: MatMul + Softmax

Ver. A) MatMul w/ psum in memory + Softmax = **1.581.056 LOAD + 528.384 STORE**

Ver. B) MatMul w/ psum in local reg + Softmax = **1.056.768 LOAD + 8.192 STORE**

II.c) Compute the total number of elementary arithmetic operations needed to compute A, assuming that FMA operations count as 2 and complex (exponentiation, division) operations count as 10.

MatMul: $64 * 64 * 128 * 1 \text{ FMA} = 1.048.576 \text{ ops}$

Softmax w/o storing exp in mem:

Denominator: $64 * (64 \text{ EXP} + 63 \text{ ADD})$

Normalization: $64 * 64 * (1 \text{ EXP} + 1 \text{ DIV})$

Total Softmax: $8.192 \text{ EXP} + 4.032 \text{ ADD} + 4.096 \text{ DIV} = 126.912 \text{ ops}$

Softmax w/ storing exp in mem:

Denominator: $64 * (64 \text{ EXP} + 63 \text{ ADD})$

Normalization: $64 * 64 * 1 \text{ DIV}$

Total Softmax: $4.096 \text{ EXP} + 4.032 \text{ ADD} + 4.096 \text{ DIV} = 85.952 \text{ ops}$

Total: MatMul + Softmax

Ver. A) MatMul + Softmax w/o storing exp in mem: **1.175.488 ops**

Ver. B) MatMul + Softmax w/ storing exp in mem: **1.134.528 ops**

EXERCISE III

A parallel cluster of RISC-V processors is executing matrix multiplication followed by softmax of matrices A (32x16) and B (16x16). Assume that the matrix multiplication computation is parallelized assigning rows of the output matrix C to the processors. Following the matrix multiplication, the softmax is parallelized again assigning rows of the output matrix to processors. Assume a barrier between the two kernels with negligible overhead. The cluster has 12 processors. Elements of A, B, C are 16bit FP numbers

III.a) Compute the L1 minimal shared memory size that is needed to store the complete matrices, and the total number of MACs needed to complete the matrix multiplication.

III.b) Compute the number of rows to be assigned to each core for the two kernels, trying to minimize the workload unbalance.

III.c) Compute the speedup with respect to sequential computation. Assume 10% of stall cycles due to banking conflicts in the parallel execution.

$$\begin{aligned}\text{Speed up} &= n_{\text{rows}} * n_{\text{kernel}} / [n_{\text{kernel}} * \text{max_number_rows_core} * (1 + \text{stalls})] = \\ &= 32 * 2 / [2 * 3 * (1 + 0.1)] = 9.7 \times\end{aligned}$$